

- PETER ARNDT, HUGO MARIANO, AND DARLLAN PINTO, *Craig interpolation from Horn semantics*.

Computer Science at the Faculty of Mathematics and Natural Sciences, Heinrich-Heine University Düsseldorf, Germany.

E-mail: `peter.arndt@hhu.de`.

Department of Mathematics at Institute of Mathematics, Statistics, and Computer Science, University of São Paulo, Brazil.

E-mail: `hugomar@ime.usp.br`.

Department of Mathematics at Institute of Mathematics, Statistics, Federal University of Bahia, Brazil.

E-mail: `darllan@ufba.br`.

In this work we consider semantics of a logic in a class of first order structures axiomatized by universal Horn sentences, a *Horn class*. We give conditions on such a semantics which ensure that an amalgamation property for the Horn class implies Craig interpolation for the logic. This generalizes the well-known result that for an algebraizable logic the amalgamation property for the associated class of algebras implies Craig interpolation.

Our main technical device is the notion of κ -filter pair ([1], [2], [3]): If κ is a regular cardinal, a κ -filter pair is a functor $G: \Sigma\text{-Str} \rightarrow \kappa\text{-AlgLat}$ together with a natural transformation to the power set functor $i: G \rightarrow \wp$, which objectwise preserves infima and κ -directed suprema.

For a κ -Horn theory $\mathbb{T}$, we define a lattice of atomic Horn formulas: For a Σ -structure A let

- $G^{\mathbb{T}}(A) := \{(\theta, S) \mid \theta \text{ is a } \Sigma\text{-congruence on } A \text{ and } S \text{ an interpretation of } \mathfrak{R} \text{ on } A/\theta \text{ s.t. the resulting } \Sigma^+\text{-structure on } A/\theta \text{ is a } \mathbb{T}\text{-model}\}$
- We define an order on $G^{\mathbb{T}}(A)$ by declaring $(\theta, S) \leq (\theta', S')$ iff $\theta \subseteq \theta'$ and the induced quotient map $q_{\theta\theta'}: A/\theta \rightarrow A/\theta'$ is a homomorphism of Σ^+ -structures for the interpretations S, S' .

Theorem Let τ be a set of *atomic* Σ^+ -formulas with at most one free variable, such that $|\tau| < \kappa$. The collection of maps $i^{\tau} = (i_A^{\tau})_{A \in \Sigma\text{-Str}}$, defined by

$$\begin{aligned} i_A^{\tau}: G^{\mathbb{T}}(A) &\rightarrow (\mathcal{P}(A), \subseteq) \\ (\theta, S) &\mapsto \{a \in A \mid \forall \varphi(x) \in \tau: A/(\theta, S) \models \varphi(a)\} \end{aligned}$$

then $(G^{\mathbb{T}}, i^{\tau})$ is a κ -filter pair.

Using the formalism of filter pairs, we can prove a general Craig Interpolation result:

Theorem Let $(G^{\mathbb{T}}, i^{\tau})$ be a Horn semantics for a logic L . Suppose that the filter pair $(G^{\mathbb{T}}, i^{\tau})$ has the “theory lifting property”. If $\mathbf{K} := \text{Mod}(\mathbb{T})$ has the atomic amalgamation property, then the logic L associated to $(G^{\mathbb{T}}, i^{\tau})$ has the Craig entailment property.

The “theory lifting property” is a technical condition, satisfied by every filter pair presenting an equivalent Horn Semantics, but also in other cases.

In the talk we will review the notion of filter pair and explain the above results with examples.

[1] P. ARNDT, H. L. MARIANO, D. C. PINTO, *Finitary Filter Pair and propositional logics*, *South American Journal of Logics*, vol. 4 (2018), no. 2, pp. 257–280.

[2] ———, *Filter Pairs and natural extension of logics*, *Archive for Mathematical Logic*, vol 62 (2023), pp. 113–145.

[3] ———, *Congruence filter pairs, equational filter pairs and adjoints*, *Journal of Logic and Computation*, vol. 35 (2025), no. 5, exae002.